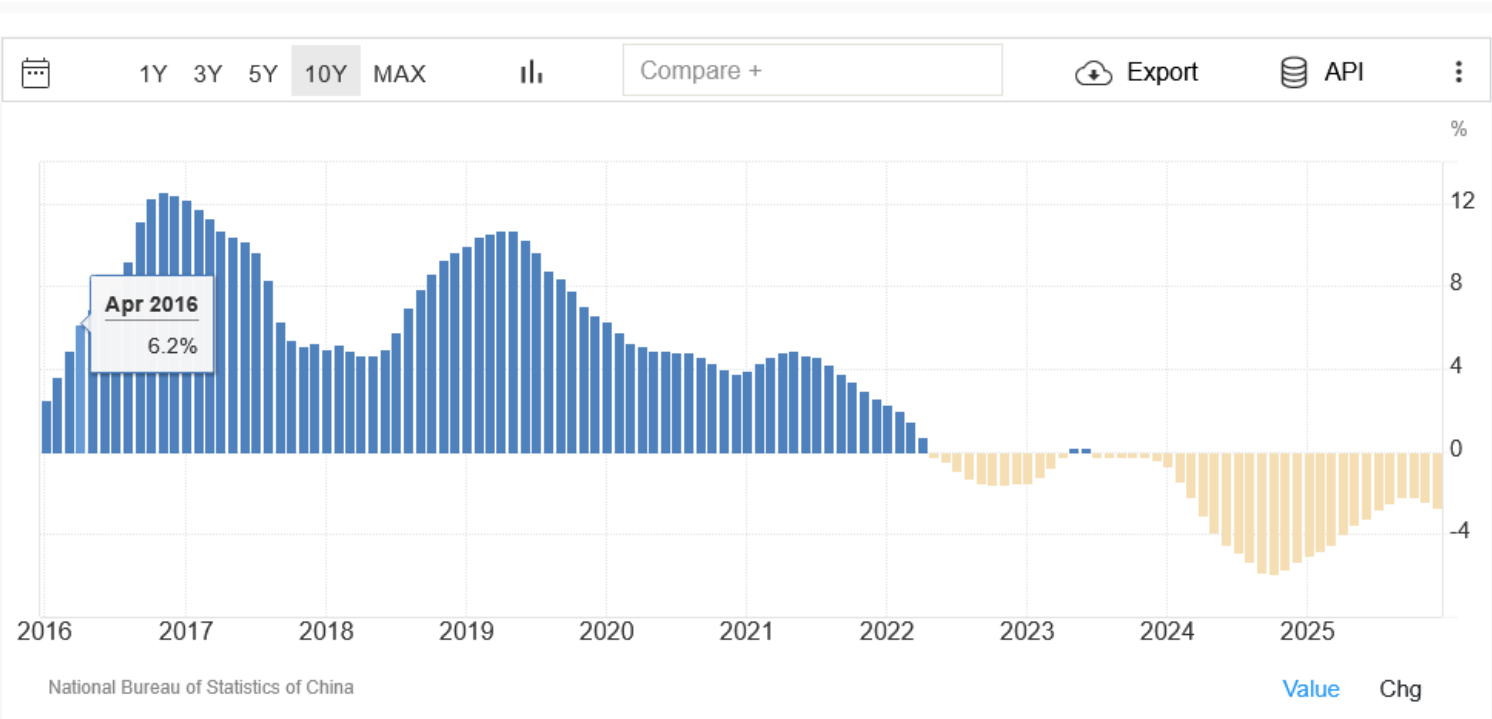
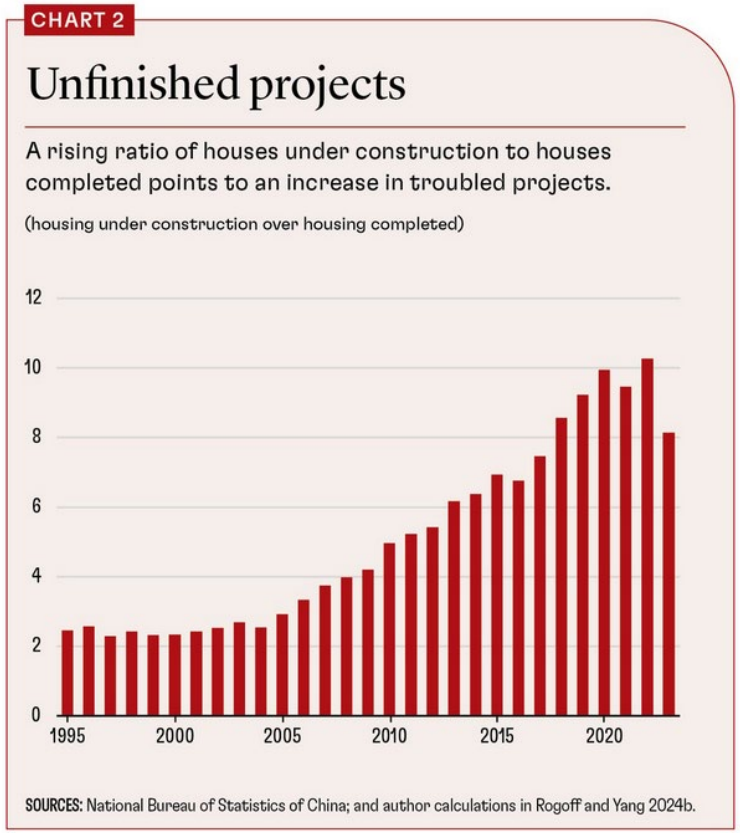
Finance in the Morning

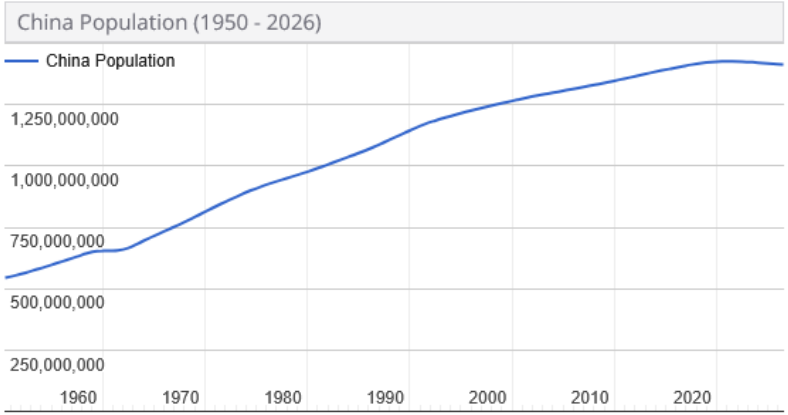
- Most important story in housing in the last two years (notice these are annual rates of change)



https://tradingeconomics.com/china/housing-index

Long term problems





In China like elsewhere finance plays a key role in every aspect of housing
Infrastructure
Land
Construction (primary market)
Resale (secondary Market)
Renting (secondary Market)

Class 13:
Real estate and Leverage

- **Problem 1: real estate**
- **Problem 2: Investment (buyer)**
- **Problem 3: Banks (lender)**
- **Sources:**
  - **Indivisibilities, spreads, transformation**
- **Options**
  - **The borrowers**
  - **The lenders**

Key conceptts

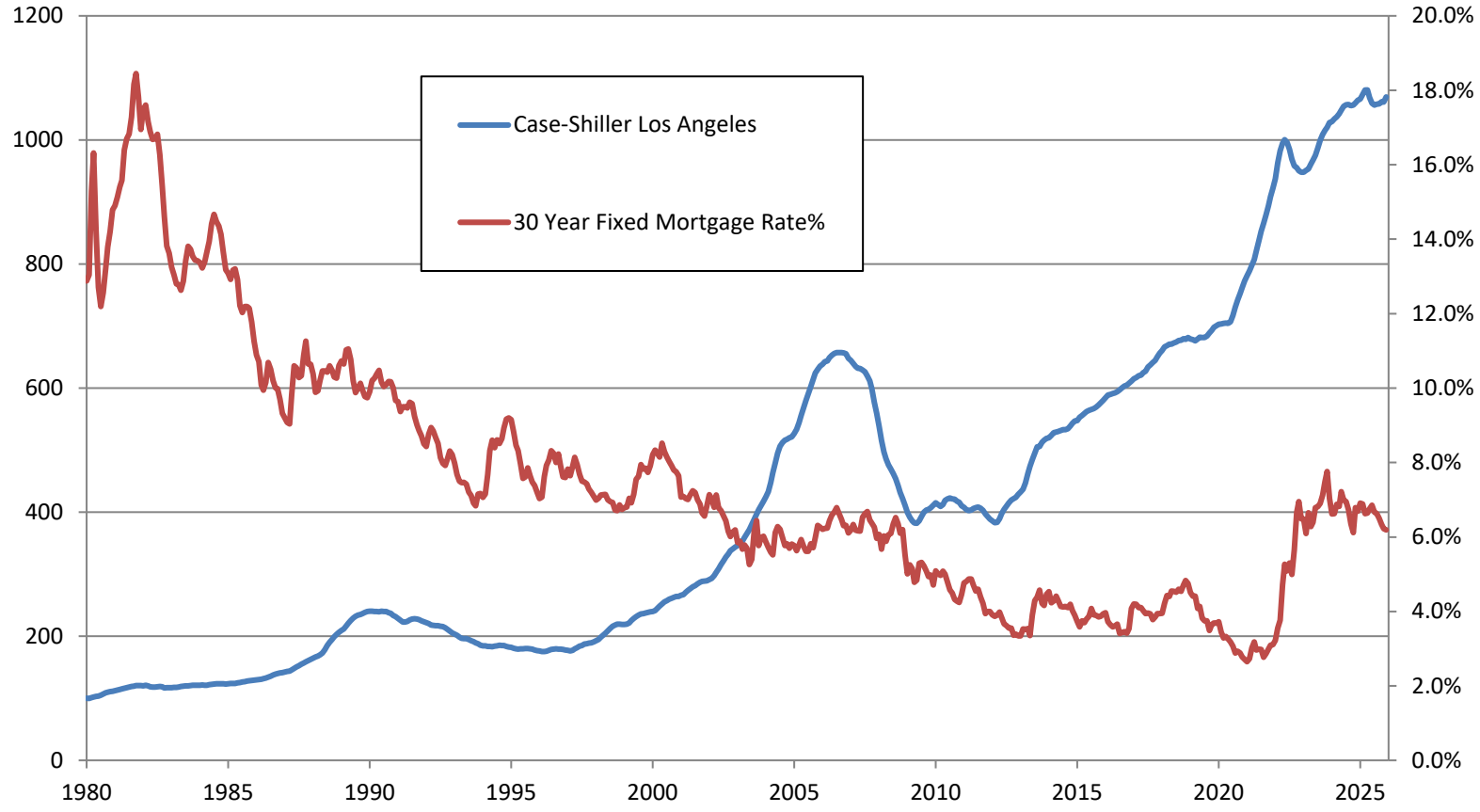
- Leverage
- Residual value (borrower’s equity)
- Value at risk (again)
- Net asset value
- Foreclosure and Limited liability
- Option

Real Estate

- Real estate huge asset class
  - By value About 50 trillion dollars
  - Very widely distributed
  - Home ownership (60%) is an important social variable
- Mortgages are $21.5 trillion
- National debt $38.5 trillion
- Equity markets $108 trillion

The housing problem

- People face a choice to rent or buy the dwelling they reside in.
- When they buy, they most often borrow (mortgage)
  - In any case they are the equity holders (and face risk)
  - Mortgage includes two options : default and early repayment.
  - Bank knows this so leverage is a decision



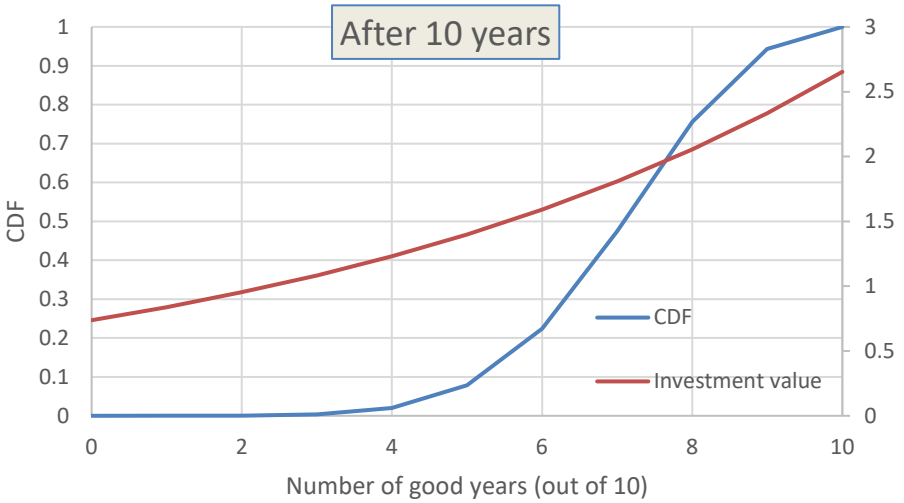
December 2026: median home for sale $926,000. (Zillow)
Average annual rate of capital appreciation since 1980 4.4%
There is variance, but its low 0.04

Setting up the binomial pb pure long investment

- 25% chance that capital gains are negative
  - And in this case drop is 7%
- 75% chance that capital gains are positive
  - Gain is 10.25

  Leave out rent

  (for now)



Problem you need the cash

Problem of housing

- Indivisible (unlike financial assets)
  - Most people cannot afford to pay cash for a home so they borrow.
  - How much they afford depends on how much they can borrow.
  - How much of a return they make (lose) on their investment depends on how much they borrow.
- So lets look at this in a specific context (Los Angeles) and assume that the only source of uncertainty is home prices.
  - Note there is a second pb which is interest risk.

The mortgage pb

- Add two things
  - On the return side housing services (you don't pay rent elsewhere). Assume net rent is 3% of value.
  - On the borrowing side you have to pay the mortgage interest rate
  - Early on rental income does not cover costs.
- Demonstrate two things
  - Leverage allows return to go up
  - Adds risk

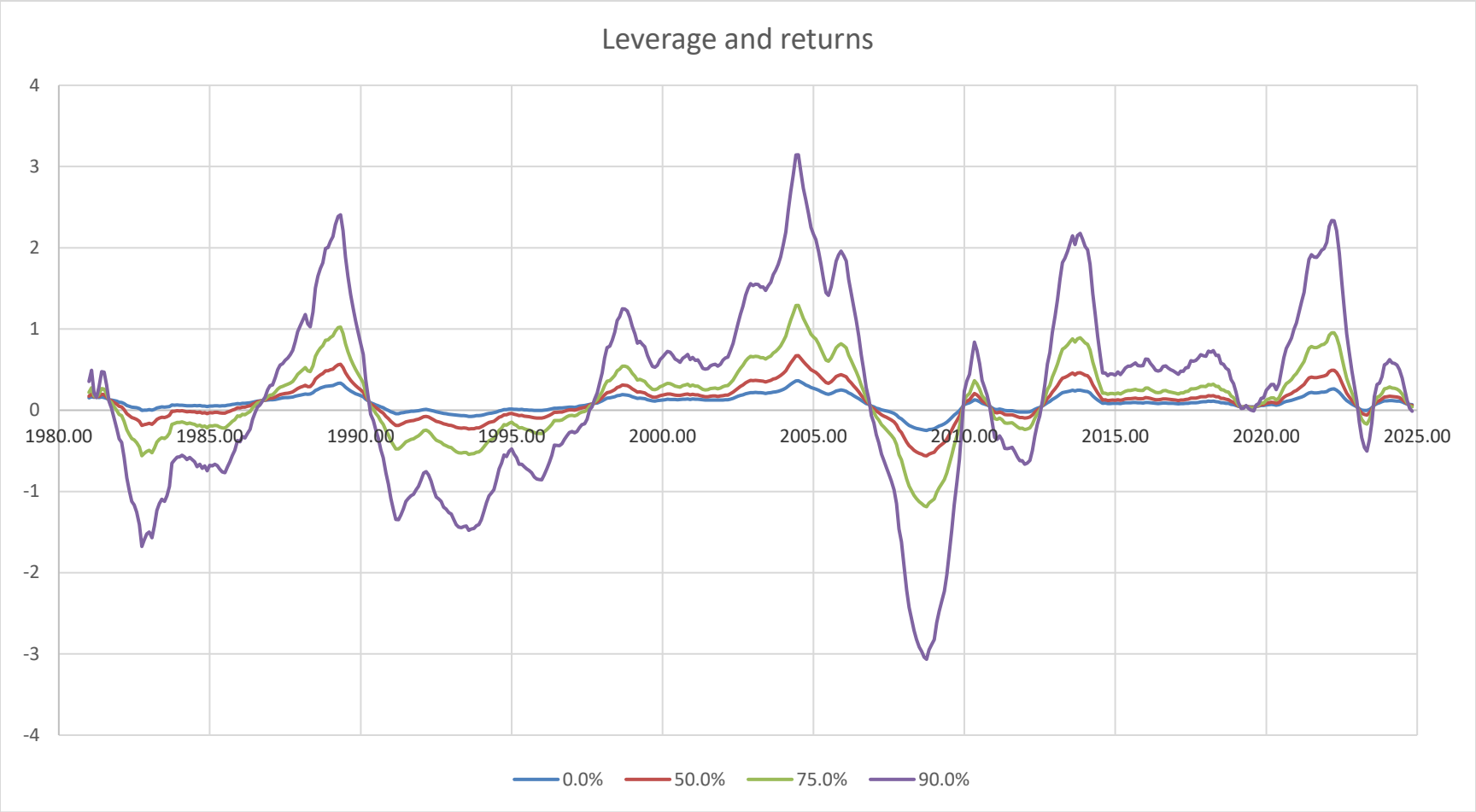
The return calculation

- Compute 1 year return
- Assume the rental income is 3% of price
- If you do not borrow $R = \frac{P_1 + (0.03P_0)}{P_0}$
- If you do borrow α of purchase price
- $R = \frac{P_1 + (0.03)P_0 - \alpha(1+r)P_0}{(1-\alpha)P_0}$
  - Leave aside taxes and self amortizing issue.

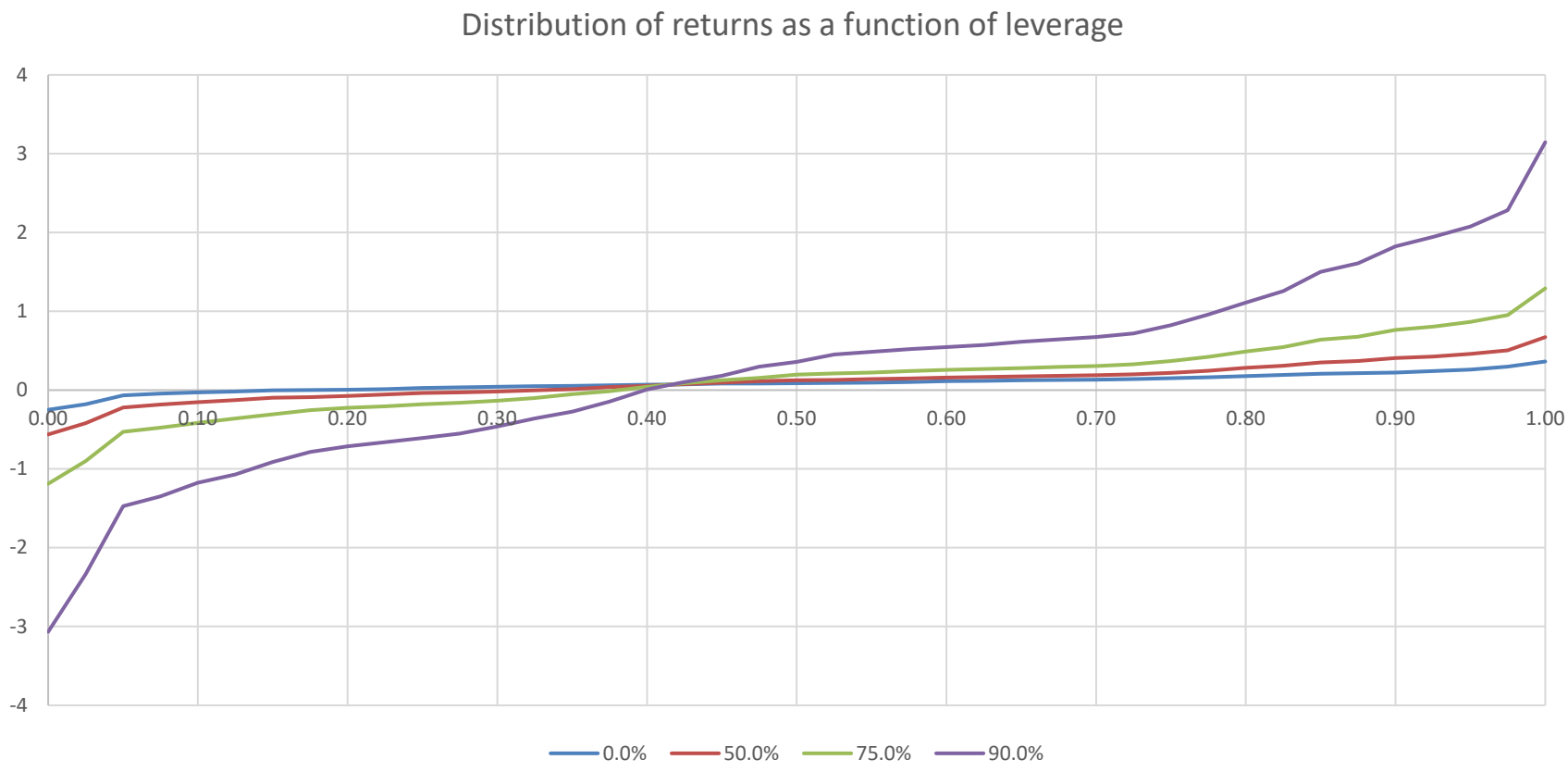
Leverage increases returns

- $\frac{P_1+(0.03)P_0-\alpha(1+r)P_0}{(1-\alpha)P_0}$
- Example 2010 April 1 year appreciation 9.7%
- With rental income r=12.7%
- Mortgage rate is 5%
- $\frac{109.7+(3)-\alpha(1.05)100}{(1-\alpha)P_0}=\frac{112.7-105\alpha}{(1-\alpha)100}$
- $\alpha = 0.5$ => $\frac{112.7-52.5}{50} = 120.4$ => R=20.4%
- $\alpha = 0.75$ => $\frac{112.7-78.75}{0.25} = 135.8$ => R=35.8%

Leverage increases variaance



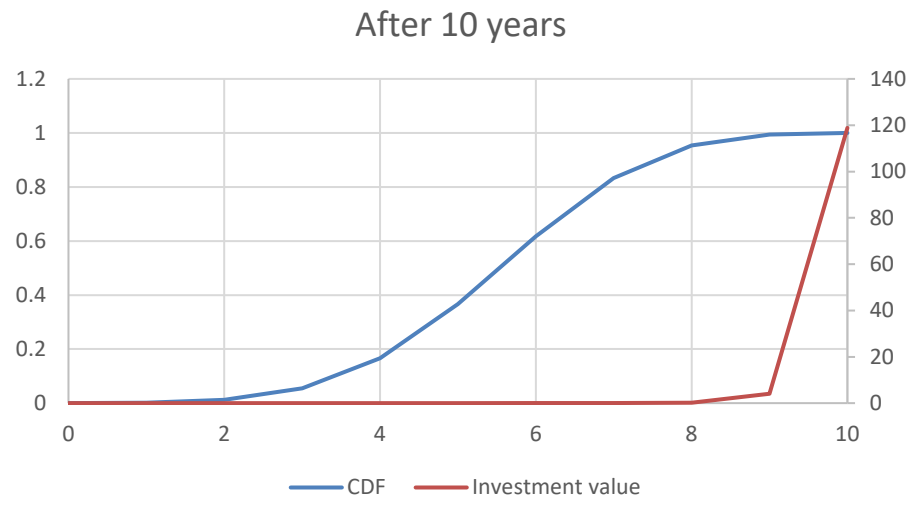
Leverage increases returns and losses



	0.0%	50.0%	75.0%	90.0%
Return	8.9%	10.4%	13.2%	21.8%

The binomial pb
90% leverage

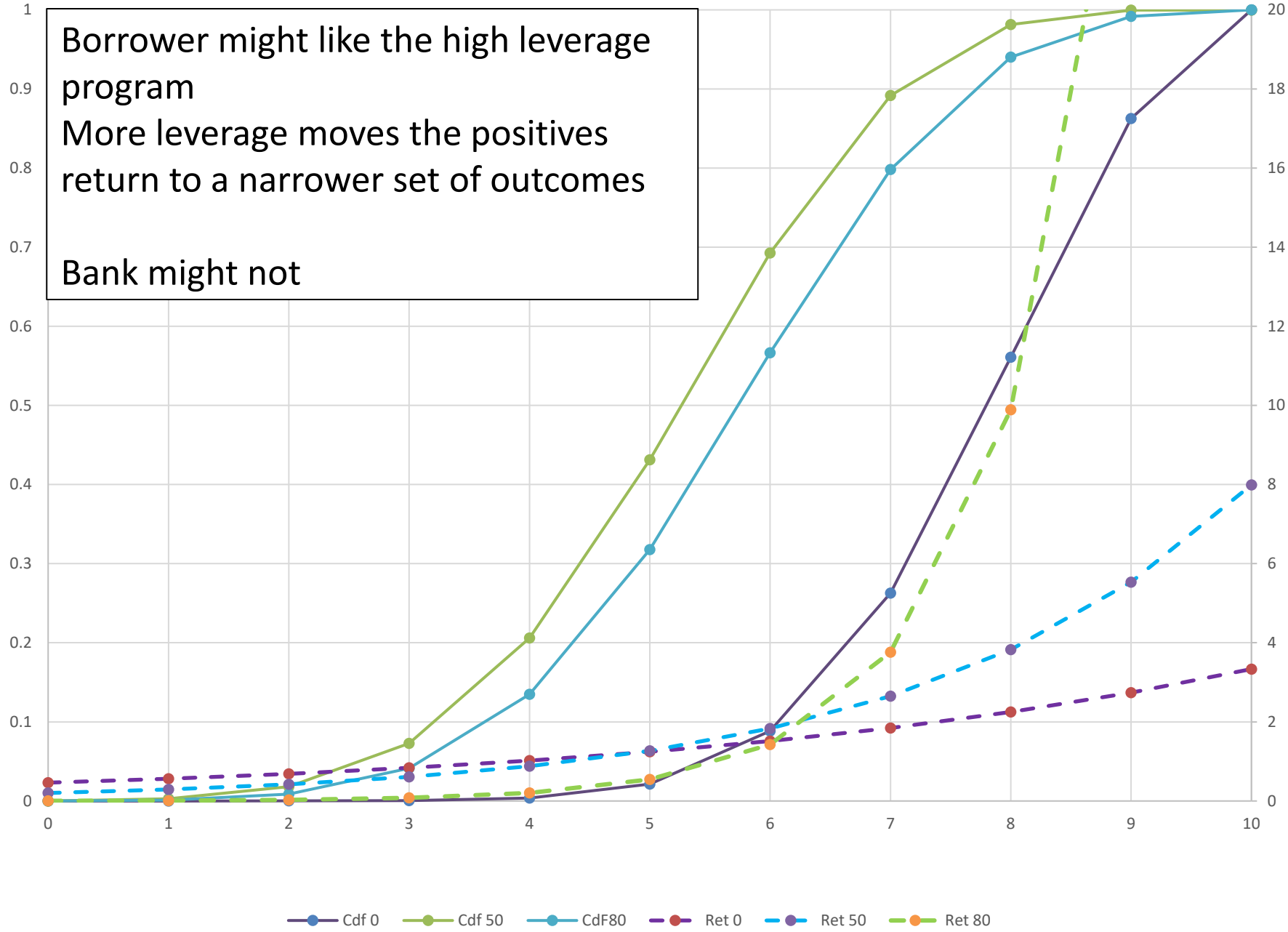
- 40% chance that returns are negative
  - And in this case drop in equity is -93% (0.07)
- 60% chance that capital gains are positive
  - Equity Gain is 103



Problem you are underwater (NAV=0) often

Adding details

- No leverage (but rental income)
  - P_d=0.18 R_d=-0.074 P_u=0.82 R_u=12.8
- Leverage 50% (the 19th century rule)
  - P_d=0.33 R_d=-0.148 P_u=0.67 R_u=23.1
- Leverage 80% (conforming mortgage)
  - P_d=0.38 R_d=-0.42 P_u=0.58 R_u=52.6


Borrower might like the high leverage program
More leverage moves the positives return to a narrower set of outcomes
Bank might not
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
20
18
16
14
12
10
8
6
4
2
0
0
1
2
3
4
5
6
7
8
9
10
Cdf 0
Cdf 50
CdF80
Ret 0
Ret 50
Ret 80

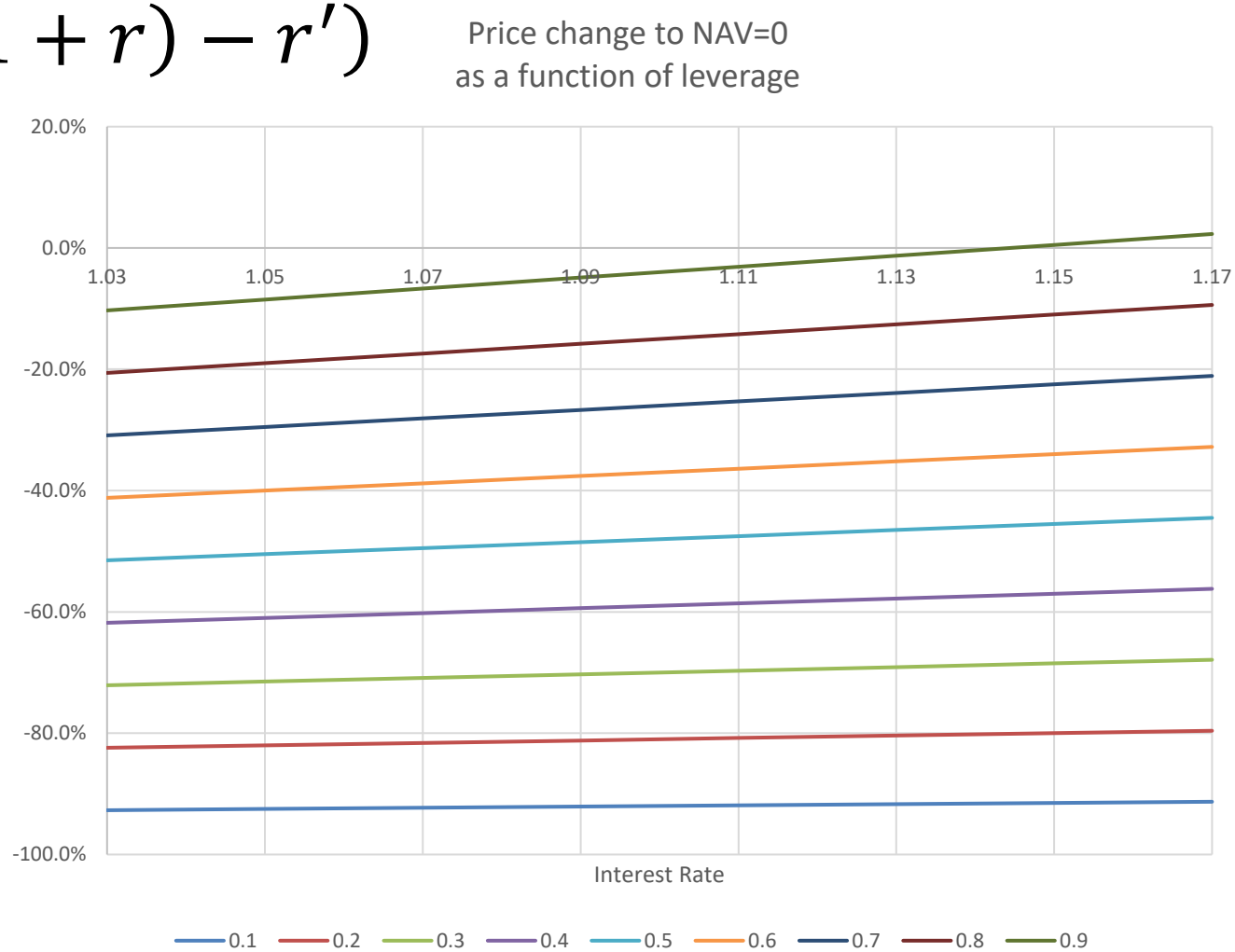
Mortgage as an option

- A mortgage is a collateralized loan.
  - Borrower can opt out by surrendering the collateral
  - The lender can take possession in case of non-performance
- Step away from homeownership and consider a levered investment in real estate
  - The collateral clause is really an option
  - So investor will step away from the loan of NAV is negative

From return to equity position

- $R = \frac{P_1 + (r\prime)P_0 - \alpha(1+r)P_0}{(1-\alpha)P_0}$
- Time 0 invest $(1-\alpha)\,P_0$
- Equity at time t+1 residual value of investment
  - Assets=house +rent
  - Liabilities Loan +interest
- $E = P_1 + r'P_0 - \alpha(1+r)P_0$
- If α=0 E= $P_1 + (0.03)P_0$
- Equity falls as leverage increases
- What is the largest drop in price such that equity is positive

- $P_1 + r'P_0 - \alpha(1+r)P_0$=0
- $P_1 - (\alpha(1+r) - r')P_0 = 0$
- $\frac{P_1}{P_0} = (\alpha(1+r) - r')$



The lender's problem

- When leverage is high enough, or the asset is sufficiently volatile.
    - Value of the long position can fall enough that it is less than the short position
    - Liquidation value is negative
  - Borrower has limited liability
  - Lender then owns the investment and faces losses.
- What to do?

In real estate as investments

- Limit leverage
  - Historically to 50% of Value.
  - The set of occurrences when prices fall by 50% is quite small
  - Now between 80 and 90%
- Out-source risk: Fannie Mae for conforming mortgages
- Charge risk premia
  - ok when interest rates are low,
  - Not good when interest rates are high

More solutions

- Force borrowers to build equity
  - Self amortizing vs ballon payments
- Diversify across vintages
  - Mortgages that have been outstanding for 3-4 years are much safer than new ones.
- Now face a different risk
  - Early repayment (interest rate risks)

Returns and foreclosure

- Returns are correlated with foreclosures but negative returns do not imply foreclosure
- Negative returns imply you are “under water”
- You can still decide to pay.
  - Because you love your house
  - Because you expect the market to recover
- The questions is what will your lender do?

Next time
bank regulation

Why regulate

How to regulate